

The Knowledge Lifecycle of Large Language Models

A model holds two memories. When they disagree, training usually wins.

Acquire

Store

Retrieve

Update

Forget



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Start with a question

You paste a document into a model's prompt.

The document contradicts what the model learned.

Which one wins?

Most of us assume the prompt wins. That assumption is what this talk tests.

The field is fragmented

- Five research communities each own one part of how a model handles a fact:
 - retrieval-augmented generation · knowledge editing
 - continual learning · context engineering · agent memory
- Each has its own benchmarks. Each passes them.
- A deployed system can pass all five and still fail, because nothing tests what happens **between** them.

Across 20 representative papers, median coverage is **two** of five stages.

The framework: five stages

Stage	What happens to the fact	Studied as
■ Acquire	Training compresses a corpus into weights	Pre-training, fine-tuning
■ Store	Weights, an external index, or both	Parametric memory, vector stores
■ Retrieve	Attention recalls, or a pipeline fetches	Retrieval-augmented generation
■ Update	The world changes; copies must follow	Knowledge editing, continual learning
■ Forget	Removed on purpose, or lost by accident	Unlearning, catastrophic forgetting

The failure in this talk sits between **Retrieve** and **Update**.

The probe

- Take a fact the model certainly holds: the capital of France.
- Put a document in the prompt asserting a plausible substitute: *the capital was relocated to Lyon.*
- Now the conflict exists **by construction**, for every model, whatever it was trained on and whenever.

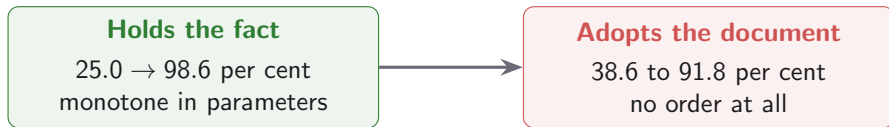


The measurement

Model	Params	Holds it	Adopts the doc
GPT-2	124M	25.0%	50.0%
GPT-2 medium	355M	54.2%	43.6%
Qwen2.5-0.5B	494M	68.1%	91.8%
GPT-2 large	774M	79.2%	75.4%
TinyLlama-1.1B	1100M	97.2%	38.6%
Qwen2.5-1.5B	1540M	98.6%	87.3%

- **864 forward passes.** Six models, 24 probes, three phrasings, with and without the document.
- Restricted to the **304** pairs where the model demonstrably held the fact, so a conflict actually existed.

Scale buys knowledge, not deference



TinyLlama holds the fact **97.2%** of the time and defers **least** of the six.
Qwen2.5-0.5B, a third its size, defers **91.8%**.

The wording moves it as much as the model does

- Each probe is asked three ways, same question, different words.
- Across the three, the adoption rate has a median spread of **47.9** percentage points and a maximum of **87.5**.
- Qwen2.5-0.5B runs from **100.0%** to **12.5%** on the *same 24 facts* with the *same document*.

A single-phrasing result about knowledge conflict is reporting a property of the sentence
at least as much as a property of the model.

Probability is not an answer

- Probability mass is spread over tens of thousands of tokens. An answer has only to be **ranked first**, not to be large.
- The in-context answer carries more than 0.10 in **332** of the 432 conditions with the document present.
- In **69** of those, **20.8%**, it is *still not* what the model says.

Recording the rank costs **one comparison** against the maximum, and it is the only quantity that matches what a deployed system emits.

A correction to our own earlier report

- An earlier version of this work reported that **no model** answers a drug-withdrawal probe correctly under any phrasing.
- **That claim was false.** The script behind it recorded only the *surprisal* of the expected answer, never its rank, so it could not have tested the claim it was cited for.
- Re-run with rank recorded: **three of the six** rank the corrected answer first under two of three phrasings.
- A second defect was structural. Those probes were real-world updates, and the withdrawal predates every model's training data, so **no conflict existed to resist.**

What is measured, and what is proposed

Measured

864 passes, six models,
deterministic,
reproducible on a laptop

Not claimed

Which factor drives it:
family, tuning and scale
vary together here

Every figure here is **generated** from the measurement, never typed.
The earlier version quoted a number its own code had never produced.

Check it yourself

- Every number in this talk comes from **one deterministic script**. No sampling, identical digits on any machine.
- Six open models from 124M to 1.5B, float32 on CPU, so the digits do not depend on an accelerator's arithmetic.
- The demonstration runs GPT-2 **inside your browser**. Nothing you type leaves your laptop.

Paper and code github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE

Live demonstration huggingface.co/spaces/ameythakur/llm-knowledge-lifecycle

What comes next

- **Which factor actually drives deference?** Family, instruction tuning and scale vary together across these six models, so the negative claim is supported and the positive one is not. Holding two fixed is the next experiment.
- **Does it hold above 1.5B?** Unknown. The non-monotonicity is evidence about this range, and 7B and above may behave differently.
- **Does reasoning-mode inference help?** Extended test-time compute should, in principle, resolve conflicts the forward pass cannot.
- **Multimodal.** A text prior overriding a contradicting image is the same failure in another modality.

Both answered, and both answers are largely negative.

In three sentences

- Five separate research fields are five stages of **one lifecycle**, and systems fail at the boundaries nobody benchmarks.
- Whether a document beats a model's memory is **not a question of size**: holding a fact scales, deferring to a correction does not.
- And it has to be read at the **rank**, because probability and answer disagree one time in five.



Thank you

Questions



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github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE

Backup: anticipated questions

- **Why models this small?** Fully open, and the whole measurement reruns on a laptop in twenty minutes. The claim is about this range and says so; 7B and above is untested.
- **Is this just hallucination?** No. The document is placed in the prompt by construction, so nothing was lost in retrieval. What fails is the step after it.
- **Aren't counterfactual documents unfair?** They are a different case from a deployed system, where the document is true and the memory is stale. They buy freedom from the training-cutoff confound, and a model that defers here may be deferring to misinformation too. Both are in the limitations.
- **Why only capitals?** It controls for question type, at the cost of saying nothing about numeric, temporal or causal facts.
- **Why does the earlier claim appear at all?** Because it was published. Correcting it on the record is cheaper than letting someone else find it.